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Paper 1: Novel Drug-Disease Modeling Framework for Oncology Benefit–Risk Evaluation: Appl

Bibliographic Information

- **Title:** Novel Drug-Disease Modeling Framework for Oncology Benefit–Risk Evaluation: Application to Tusamitamab Ravtansine
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- **Journal:** CPT: Pharmacometrics & Systems Pharmacology, 2026, 15(2): e70190
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Paper Type: METHODOLOGY

Quick Take

This paper presents a novel drug-disease modeling framework that integrates pharmacokinetics, dose-modifying adverse events (DMAEs), tumor dynamics, and progression-free survival (PFS) through an iterative simulation process. Applied to tusamitamab ravtansine in non-squamous NSCLC, the framework accurately predicted phase III outcomes and revealed a critical trade-off between dose-dependent efficacy gains and corneal toxicity, demonstrating its utility for early-phase dose optimization.

Executive Summary

The authors develop a comprehensive drug-disease modeling (DDM) framework that explicitly models the feedback loop between safety-driven dose modifications and treatment effectiveness. The framework comprises four components: population PK, repeated time-to-event (RTTE) models for DMAE occurrence/grade/duration, a tumor growth inhibition (TGI) model with log-kill hypothesis, and a joint TS/PFS model with log-logistic baseline hazard. Using phase I data from 254 patients (88 nsq NSCLC for efficacy), the model successfully predicted phase III PFS (4.4 vs 4.2 months) and ORR (24% vs 27.8%). Dose simulations (80, 100, 120 mg/m² Q2W) showed that while ORR improved with dose, PFS gains were minimal and corneal events increased substantially, highlighting the framework's ability to quantify benefit–risk trade-offs.

Scientific Context & Motivation

Optimizing the benefit–risk balance of antibody-drug conjugates (ADCs) in oncology requires integrated frameworks that account for the interplay between dose-modifying adverse events and treatment effectiveness. Current modeling approaches often treat safety and efficacy separately, missing critical interactions such as dose reductions due to toxicity that can compromise efficacy. This work addresses the need for a unified DDM platform that can leverage early-phase data to predict late-stage outcomes and inform dose selection, particularly for ADCs where corneal and other toxicities frequently necessitate dose modifications.

✂ Methodological Snapshot

The framework integrates four components: (1) a population PK model (semi-mechanistic for ADC, DM4, MeDM4) providing time-varying exposure metrics; (2) a DMAE safety model using RTTE for occurrence (exponential distribution), logistic regression for severity grade, and linear mixed-effects for duration; (3) a TGI model (Claret log-kill) for tumor size dynamics with instantaneous ADC concentration as exposure driver; and (4) a joint TS/PFS model with log-logistic baseline hazard and instantaneous TS slope as link function. The key innovation is the iterative simulation process that cycles through PK simulation, DMAE generation, dose adjustment, and efficacy endpoint generation, explicitly modeling the feedback loop where AEs trigger dose modifications that alter subsequent exposure and outcomes. Model estimation uses SAEM in Monolix 2023R1, with covariate selection via COSSAC or stepwise covariate modeling.

☒ Statistical Framework

The framework assumes: (1) DMAE inter-arrival times follow exponential distributions with rate dependent on time-varying exposure (ADC Cmax); (2) DMAE severity grades follow a logistic model; (3) DMAE durations follow lognormal distributions; (4) tumor size dynamics follow the Claret TGI model with log-kill hypothesis, assuming exponential growth, linear drug-induced kill, and exponential resistance; (5) PFS baseline hazard follows a log-logistic distribution; (6) covariates enter the hazard multiplicatively (proportional hazards); (7) the link between tumor dynamics and PFS hazard is linear through the instantaneous TS slope. The joint model assumes that the TS slope captures all treatment effect on PFS (i.e., no direct exposure effect beyond tumor dynamics). Estimation uses the SAEM algorithm, which provides maximum likelihood estimates under the nonlinear mixed-effects framework. Model selection uses corrected Bayesian Information Criterion (BICc).

Estimator Behavior

All parameters were precisely estimated with relative standard errors (RSE) below 37% for fixed effects and below 36% for random effects in the final models. The safety model showed substantial interindividual variability in corneal DMAE rate ($\omega = 0.86$) and grade logit ($\omega = 2.45$). The efficacy model showed moderate to high interindividual variability in tumor growth parameters (ω for $k_{ge} = 0.911$, $k_{kill} = 0.903$, $\lambda = 1.306$). The SAEM algorithm converged successfully, and standard errors were obtained via asymptotic approximation. Model evaluation via VPCs (accounting for dropout) showed satisfactory calibration for both safety and efficacy endpoints, though corneal events were underpredicted in the phase III external validation.

Validation Design

Internal validation used VPCs for safety endpoints (mean DMAE frequency, grade distribution, duration) and efficacy endpoints (TS percentiles, KM PFS curves) in the phase I nsq NSCLC subset (n=88). VPCs accounted for two competing dropout mechanisms: treatment discontinuation due to DMAEs and dropout due to progression/death. External validation compared model predictions

(500 trials of 225 patients using phase III baseline characteristics) against observed phase III outcomes for the 100 mg/m² Q2W arm: PFS (predicted median 4.4 months vs observed 4.2), ORR (24% vs 27.8%), corneal DMAE rate (8% vs 19.6%), non-corneal DMAE rate (25% vs 35.1%). No formal calibration or cross-validation was reported.

Applicability Boundaries

The framework is most applicable when: (1) early-phase data include multiple dose levels with sufficient PK and AE timing information; (2) the target population's baseline characteristics are well-characterized (or can be generated via copula methods); (3) the drug exhibits dose-modifying AEs that affect exposure; (4) tumor size is a meaningful biomarker for the clinical endpoint. Limitations: (1) the specific model structures (exponential DMAE, Claret TGI, log-logistic PFS) may not generalize to all drugs or indications; (2) requires rich longitudinal data (multiple TS measurements, AE records); (3) safety predictions may be biased if AE monitoring differs between early and late phases; (4) the framework does not model overall survival or competing risks beyond progression; (5) computational cost of iterative simulations may be high for large trial simulations.

Comparison to Alternatives

Compared to separate safety and efficacy models, this framework captures the feedback loop between AEs and treatment effectiveness, enabling more realistic simulations of dose modifications. Compared to multistate Markov models for AEs (Niebecker et al., Xu et al.), the RTTE approach avoids artificial time binning, handles multiple events per patient naturally, and directly incorporates time-varying exposure. Compared to logistic regression for weekly AE probability (Fostvedt et al.), RTTE provides a more flexible continuous-time framework. For efficacy, the joint TS/PFS model with TS slope as link function is preferred over separate TS and PFS models because it reduces bias and improves efficiency (Desmée et al., Thai et al.). The framework's main advantage is its comprehensive integration; its main disadvantage is increased complexity and the need for rich data.

Implementation Guidance

The framework was implemented using Monolix 2023R1 for model estimation (SAEM algorithm) and custom simulation scripts (likely R or MATLAB) for the iterative simulation process. Practical tips: (1) start with separate PK, safety, and efficacy models before integrating; (2) use COSSAC or SCM for covariate selection to avoid overfitting; (3) carefully define DMAE categories based on clinical relevance and dose modification rules; (4) validate dropout mechanisms using VPCs that account for competing risks; (5) for phase III prediction, ensure baseline characteristics match the target population or use virtual patient generation. Computational cost: 500 trials of 225 patients with iterative simulation is feasible on a standard workstation but may require several hours. The framework is generalizable to other drugs but requires re-estimation of all model components.

☒ Key Findings

The integrated DDM framework accurately predicted phase III outcomes for the tusamitamab ravtansine arm: median PFS 4.4 months (90% PI: 3.9–4.9) vs observed 4.2 months; ORR 24% (90% PI: 20%–29%) vs 27.8%. Safety predictions were slightly underpredicted (corneal DMAEs: 8% predicted vs 19.6% observed). Dose simulations revealed a critical trade-off: increasing dose from 80 to 120 mg/m² Q2W improved ORR from 20% to 28% but increased corneal events from 5% to 12%, while median PFS only improved marginally (4.3 to 4.6 months). The instantaneous ADC concentration was the best exposure driver for both safety (corneal DMAE risk) and efficacy (tumor shrinkage), and the instantaneous tumor size slope was the best predictor of PFS.

Strengths & Limitations

Strengths

- Comprehensive integration of PK, safety, tumor dynamics, and survival in a single iterative simulation framework
- Explicit modeling of the feedback loop between DMAEs, dose modifications, and treatment outcomes
- Use of RTTE models for DMAEs avoids artificial time binning and handles multiple events per patient
- Joint TS/PFS modeling reduces bias and improves efficiency compared to separate analyses
- Successful external validation by predicting phase III outcomes from phase I data alone
- Clear demonstration of dose–response trade-offs enabling quantitative benefit–risk assessment

Limitations (Acknowledged by Authors)

- Restricted dose variability for efficacy evaluations (most nsq NSCLC patients received 100 mg/m² Q2W)
- Prospective application requires baseline patient characteristics representative of the target phase III population
- Focus on DMAEs rather than all AEs may limit generalizability of safety assessment

Limitations (Expert Review)

- Safety predictions underpredicted corneal events (8% vs 19.6% observed), suggesting potential model misspecification or differences in AE reporting between phases
- The framework was demonstrated on a single ADC; generalizability to other drug classes or mechanisms is untested
- Computational complexity of iterative simulations may be a barrier for routine use
- Covariate selection used COSSAC/SCM without explicit discussion of multiplicity correction
- No formal sensitivity analysis on key structural assumptions (e.g., exponential DMAE inter-arrival times, log-logistic PFS baseline)

Generalizability

The framework is designed to be drug-agnostic and applicable to any oncology therapy where dose-modifying adverse events and tumor dynamics are relevant. However, the specific model

structures (e.g., exponential DMAE occurrence, Claret TGI model) may need re-evaluation for different mechanisms or toxicity profiles. The requirement for rich phase I data with multiple dose levels and detailed AE timing may limit application to early-stage programs with limited data.

Key Equations

Tumor Growth Inhibition Model (Claret log-kill)

$$dT S dt = TS \times Kg - Ks \times EXPOSURE(t) \times \exp(-R \times t)$$

Describes tumor size dynamics under treatment: Kg is growth rate, Ks is drug-induced killing rate, $EXPOSURE$ is drug exposure, R is acquired resistance rate.

Log-logistic Baseline Hazard for PFS

$$h_0(t) = \frac{s}{Te} \times \frac{(t/Te)^{s-1}}{1 + (t/Te)^s}$$

Baseline hazard function for progression-free survival, where s is shape parameter and Te is scale parameter.

Individual Hazard with Covariates

$$h_i(t) = h_0(t) \times \exp(\beta_w \times w_i)$$

Individual hazard incorporates baseline covariates w_i with coefficients β_w .

Joint Model Hazard with Link Function

$$h_i(t) = h_0 \times \exp(\beta_w \times w_i + \beta_{\text{link}} \times L(t))$$

Joint model linking tumor dynamics to PFS hazard via link function $L(t)$, with β_{link} quantifying the association.

Link Function: Current Tumor Size

$$L(t) = TS(t)$$

Link function using current tumor size $TS(t)$ as predictor of PFS hazard.

Link Function: Tumor Size Slope

$$L(t) = \frac{dT S}{dt}$$

Link function using instantaneous tumor size slope $\frac{dT S}{dt}$ as predictor of PFS hazard; this was the best-performing link in the case study.

Figures & Tables

- **Figure 1:** Schematic representation of the integrated drug-disease model showing the feedback loop between dose, PK, DMAEs, tumor dynamics, and PFS.

- *Significance*: Illustrates the framework’s structure and the iterative simulation process that captures safety-efficacy interplay.
 - **Figure 2**: Simulation procedure flowchart detailing the iterative steps: dose generation, PK simulation, DMAE simulation, dose adjustment, and efficacy endpoint generation.
 - *Significance*: Provides the algorithmic blueprint for implementing the framework in practice.
 - **Figure 3**: Internal evaluation VPCs for safety (mean number of DMAEs, grade distribution) and efficacy (TS percentiles, KM PFS) in nsq NSCLC patients (n=88).
 - *Significance*: Demonstrates adequate model fit accounting for competing dropout mechanisms (DMAE discontinuation and progression).
 - **Figure 4**: Simulated outcomes for 80, 100, and 120 mg/m² Q2W doses: corneal DMAE proportion, non-corneal DMAE proportion, ORR, and median PFS with 90% prediction intervals.
 - *Significance*: Key figure showing the dose-dependent trade-off: ORR improves with dose but corneal events increase substantially while PFS gains are minimal.
 - **Table 1**: Study design summary: phase I dose escalation/expansion cohorts and phase III arms with patient counts.
 - *Significance*: Documents the data sources and sample sizes used for model building and validation.
 - **Table 2**: Baseline characteristics of phase I and phase III patients, including demographics, tumor size, prior therapies, and CEACAM5 expression.
 - *Significance*: Highlights differences between phase I and III populations (e.g., fewer prior therapies in phase III) that may affect model predictions.
 - **Table 3**: Parameter estimates for the final safety and efficacy models, including fixed effects, random effects, and error model parameters.
 - *Significance*: Provides the quantitative basis for the framework, showing key covariate effects and interindividual variability.
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Code & Reproducibility Assessment

No public code repository is provided. The analysis was performed using Monolix 2023R1 (SAEM algorithm) and custom simulation scripts. Data availability is subject to Sanofi’s data-sharing policy (<https://www.clinicalstudydatarequest.com>).

Supplementary Materials

Supplementary materials include detailed model selection results, covariate analysis steps, VPCs for the full phase I safety model, and additional goodness-of-fit diagnostics. These are referenced in the main text but not provided in the extracted content.

Future Directions

Prospective validation of the framework across multiple ADCs and other oncology modalities is needed. Extensions could include modeling overall survival, incorporating biomarker-driven resistance mechanisms, and using copula-based virtual patient generation to overcome baseline

characteristic mismatches. Integration with adaptive trial design platforms could further enhance utility.

Expert Commentary

This work represents a significant step toward model-informed drug development for ADCs, where the interplay between toxicity and efficacy is particularly challenging. The iterative simulation approach that explicitly models dose modifications due to AEs is a key innovation. However, the underprediction of corneal events in the phase III prediction warrants careful investigation—it may reflect differences in AE monitoring intensity between phase I and III, or unmodeled time-varying covariates. The framework’s reliance on Monolix and custom simulation code may limit reproducibility; open-source implementation would accelerate adoption.

Bottom Line

This DDM framework provides a rigorous, quantitative approach to integrate safety and efficacy from early-phase trials, enabling model-based dose optimization and benefit–risk assessment for oncology ADCs. Practitioners should consider adopting this framework when early-phase data include multiple dose levels and detailed AE records, as it can inform go/no-go decisions and phase III trial design by predicting late-stage outcomes and revealing dose–response trade-offs.

☒ Figures

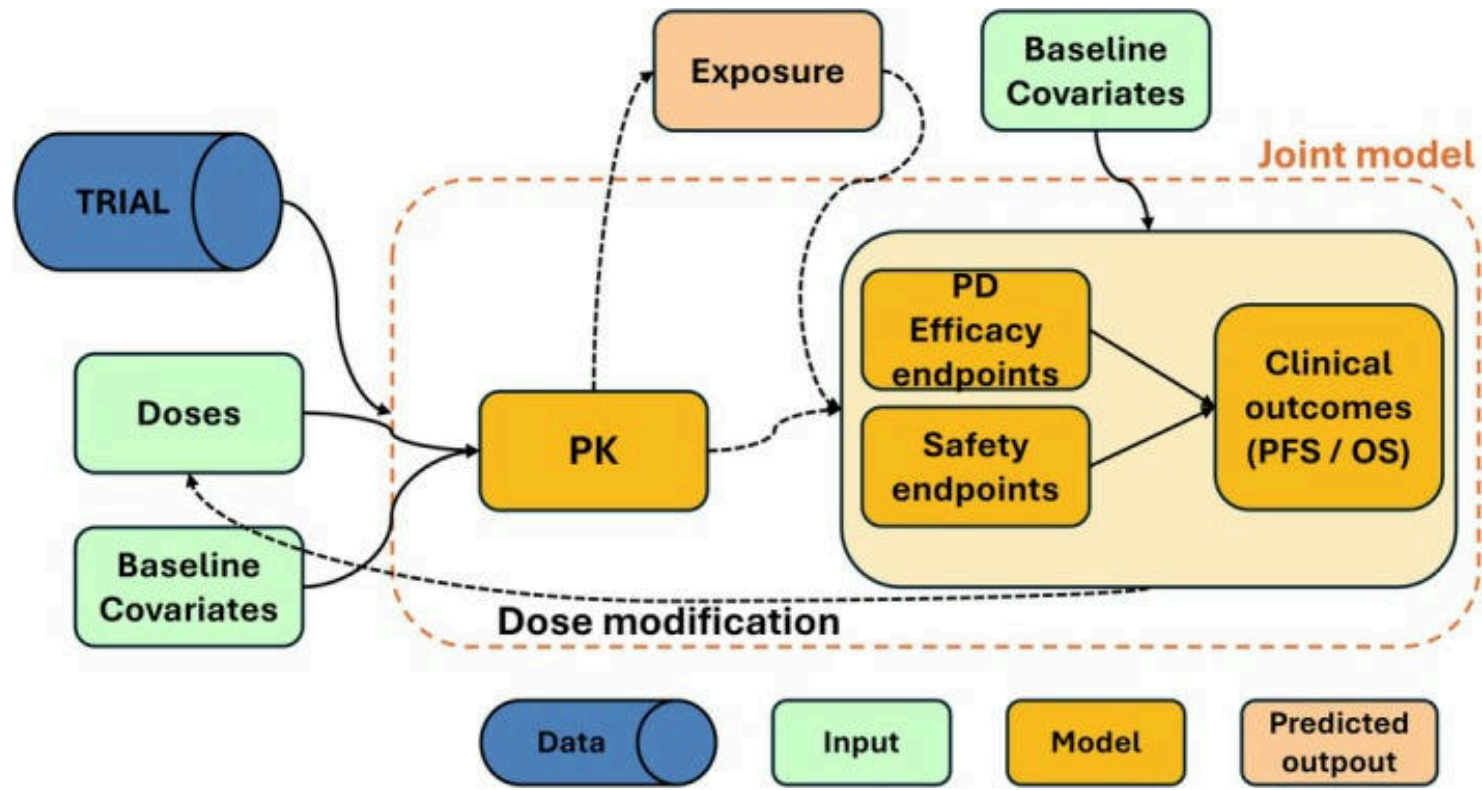


Figure 1: Schematic representation of the integrated drug disease model.

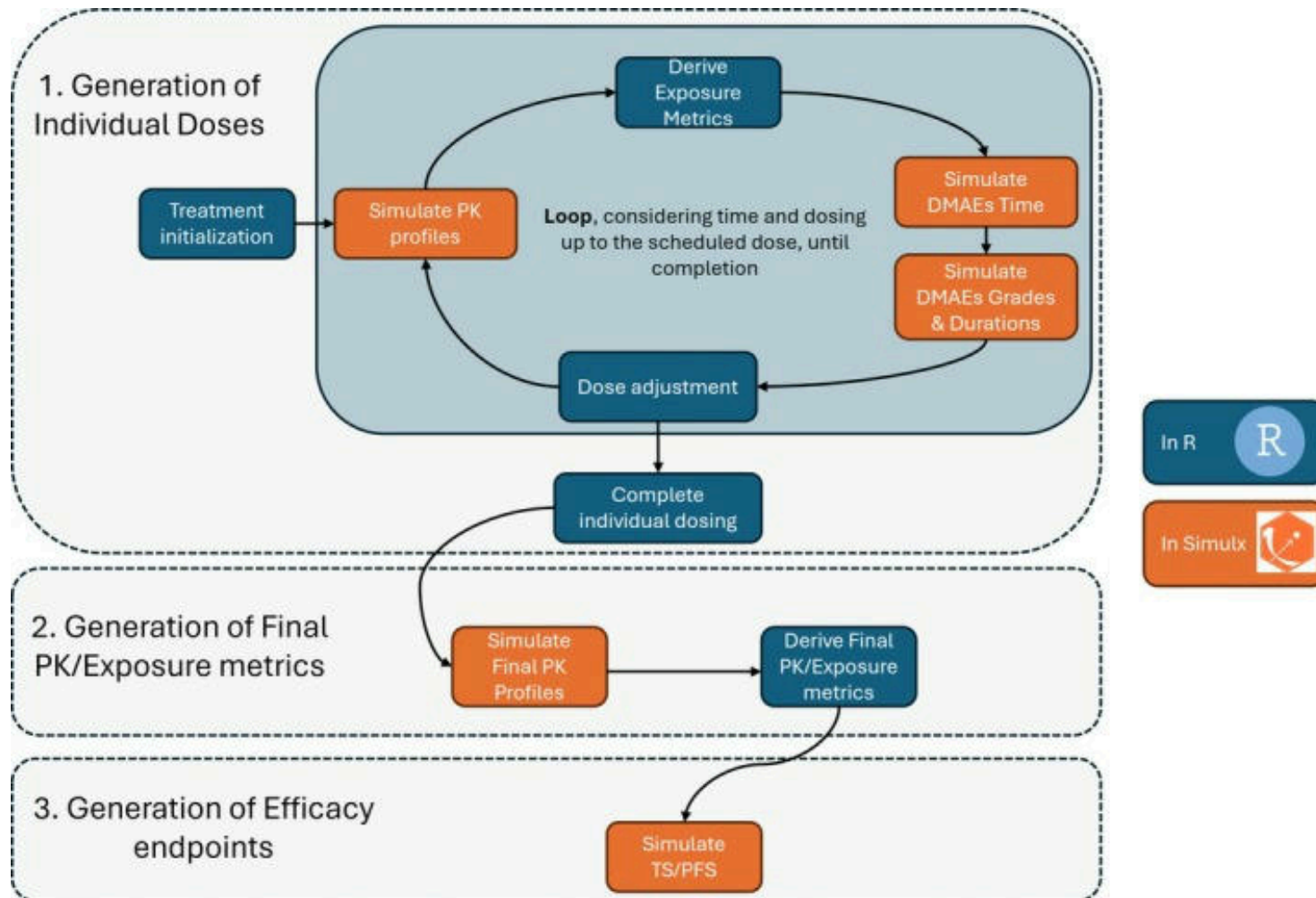


Figure 2: Simulation procedure.

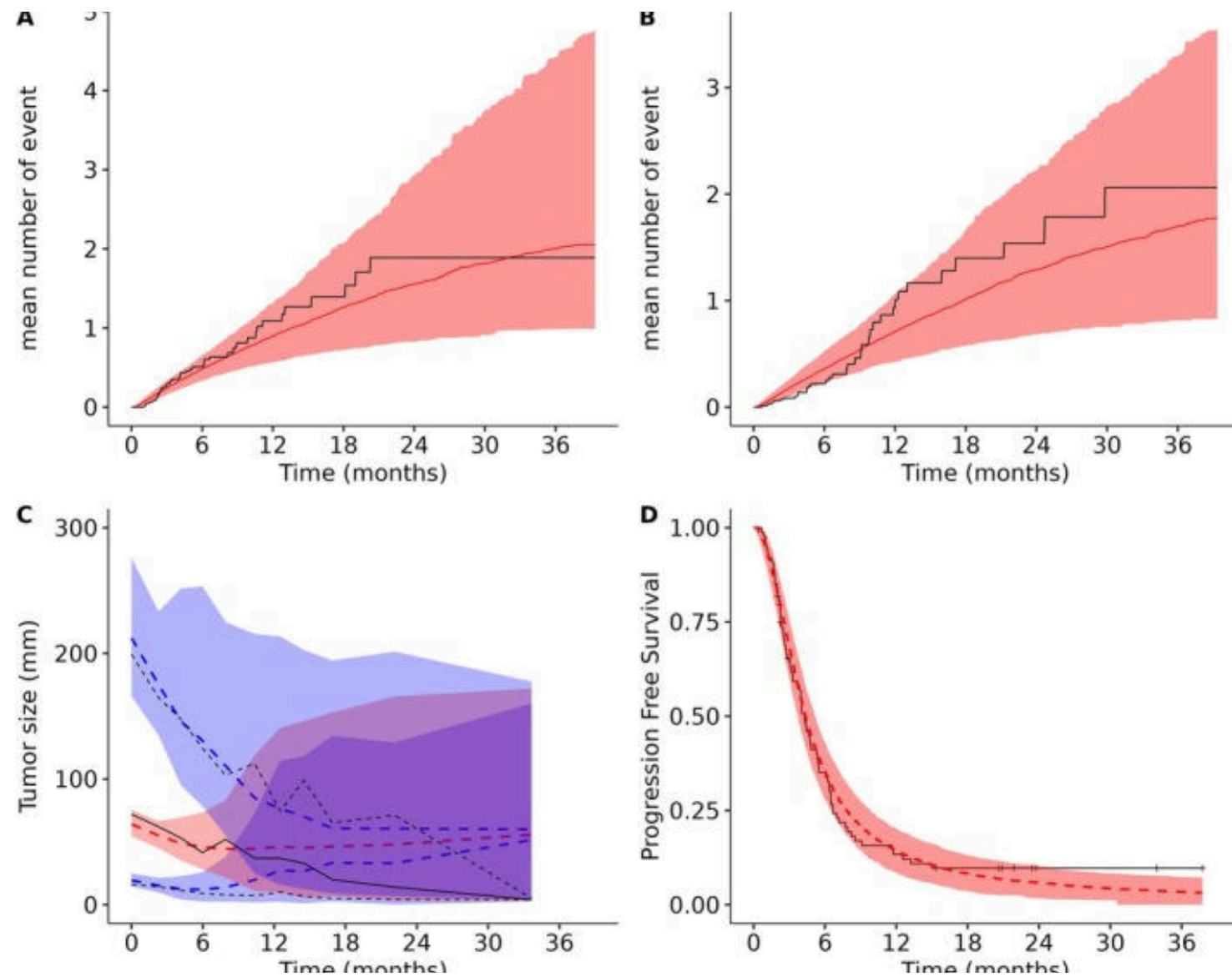


Figure 3: Internal evaluation of the model showing safety endpoints (A, B) and efficacy endpoints (C, D) based on the nsq NSCLC patients from the phase I (n= 88). Safety endpoints: VPC of th

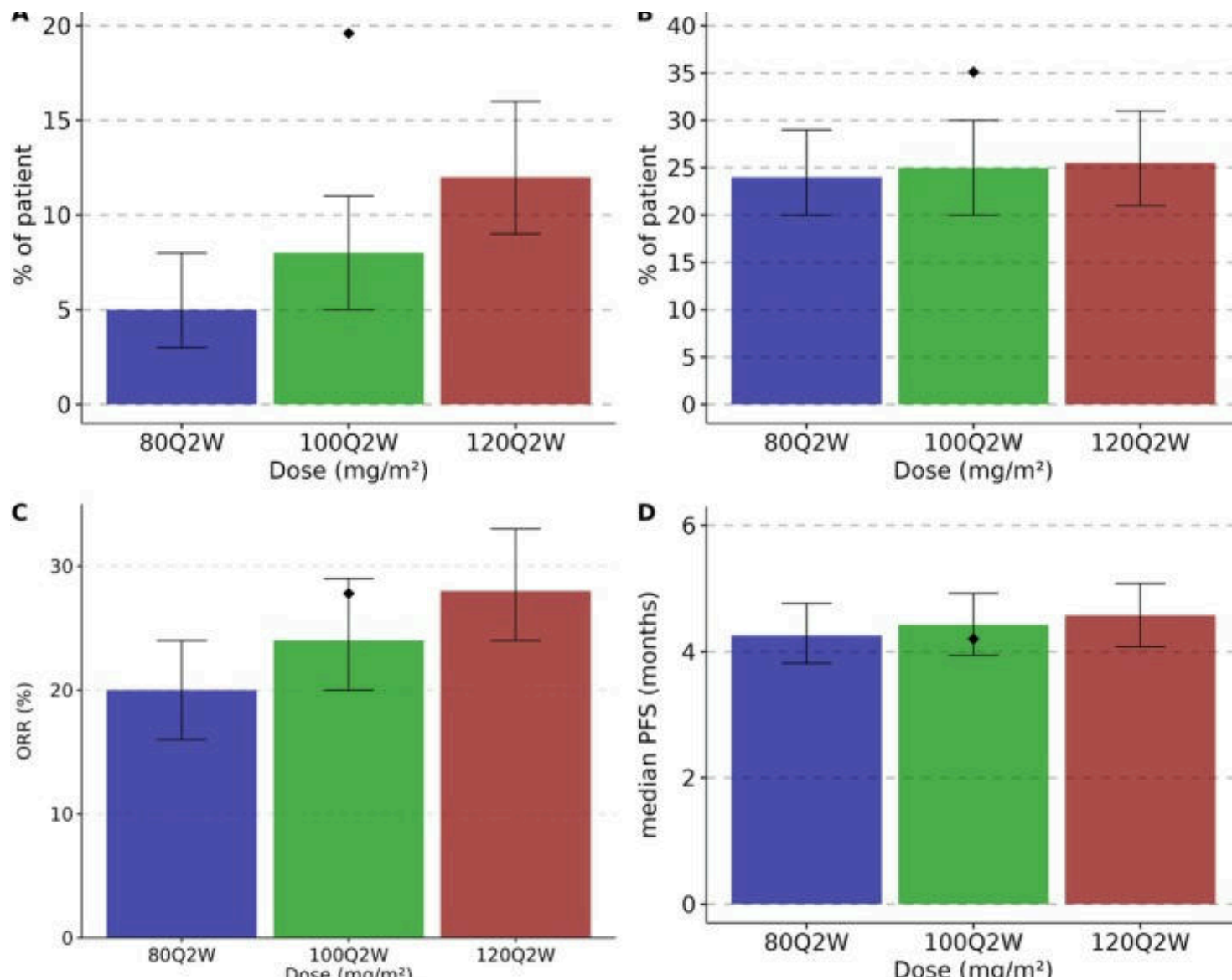


Figure 4: Simulated outcomes of different dosing regimens showing safety endpoints (A, B) and efficacy endpoints (C, D) with 90% prediction interval, for 500 trials of 225 nsq NSCLC patients